

1. Administrative & Study Identification

Full Study Title: A Multicenter, Randomized, Double-blind, Placebo-controlled, Phase 3 Study to Evaluate the Efficacy, Safety, and Tolerability of BMS-986278 in Participants With Progressive Pulmonary Fibrosis **Short Title/Acronym:** ALOFT PPF / IM027-1015 **Protocol Number:** IM027-1015 **NCT Number:** NCT06025578 **Sponsor Name:** Bristol-Myers Squibb **Investigational Product:** BMS-986278 (lysophosphatidic acid receptor 1 (LPA1) antagonist) **Phase of Development:** Phase 3 **Medical Monitor:** Head of Global Submission Management - Clinical Trials (BMS Clinical Trials Contact Center)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the efficacy, safety, and tolerability of BMS-986278 in participants with Progressive Pulmonary Fibrosis (PPF). **Secondary Objectives:** To evaluate the effect of BMS-986278 on delaying disease progression, improving or maintaining pulmonary function, and reducing mortality in PPF.

2.2 Background & Rationale There remains a large unmet medical need for a safe, well-tolerated, and effective therapy for pulmonary fibrosis. Progressive pulmonary fibrosis (PPF) is a chronic, life-threatening interstitial lung disease (ILD) characterized by damaged and scarred lung tissue. Increased lysophosphatidic acid (LPA) levels and activation of the LPA1 receptor are involved in the pathogenesis of pulmonary fibrosis. BMS-986278 is an investigational first-in-class oral LPA1 antagonist that has shown favorable safety and efficacy in Phase 2 trials, warranting this pivotal Phase 3 study.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$N\$: Up to 1,092 participants (Cohort 1: 60 participants for safety run-in; Cohort 2: up to 1,032 participants) **Number of Sites:** Multicenter, Global **Estimated Study Start:** 25-Oct-2023 **Estimated Primary Completion:** 27-Dec-2027

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 21 years. 2. Diagnosis of interstitial lung disease (ILD) with features consistent with progressive ILD within 24 months prior to screening, and $\geq 10\%$ extent of fibrosis on screening high-resolution computed tomography (HRCT). 3. If receiving background therapies (e.g., pirfenidone, nintedanib, MMF, MA, AZA, Tacrolimus, or traditional/biologic DMARDs), the participant must be on a stable dose for at least 90 days prior to screening.

4.2 Exclusion Criteria 1. Idiopathic pulmonary fibrosis with usual interstitial pneumonia (UIP) verification at screening. 2. History of stroke or transient ischemic attack within 3 months prior to screening. 3. Use of systemic corticosteroids equivalent to prednisone > 15 mg/day within 4 weeks prior to screening and during the study.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: BMS-986278 Dose 1, BMS-986278 Dose 2, or matching Placebo administered twice daily (BID). Route of Administration: Oral (by mouth). Duration of Treatment: Up to study completion (anticipated 2027 follow-up).

5.2 Concomitant & Prohibited Therapy Permitted: Background antifibrotic therapies (pirfenidone or nintedanib) and immunosuppressive therapies (MMF, AZA, tacrolimus, traditional DMARDs, biologic DMARDs, and JAK inhibitors) provided doses have been stable for ≥ 90 days prior to screening. Prohibited: Systemic corticosteroids equivalent to prednisone > 15 mg/day.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, HRCT Scan, Pulmonary Function Tests, Background Medication Verification
Baseline	Day 0	Randomization, First Dose, Vitals, Baseline Efficacy Assessment
Interim	Week [X] $\pm$ [Y]	Safety Labs, Efficacy Assessment (FVC/Spirometry), AE Monitoring
End of Study	Week [X]	Final Vital Signs, Exit Interview, IP Accountability, Final Pulmonary Function Test

(Note: Specific interim visit cadences follow standard Phase 3 ILD monitoring protocols)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Efficacy, safety, and tolerability of BMS-986278, primarily evaluated through the rate of decline in Forced Vital Capacity (FVC) or related disease

progression metrics over the specified treatment period.
Measurement Method: Standardized Pulmonary Function Testing (Spirometry) and clinical safety assessments.

7.2 Secondary & Exploratory Endpoints * Delay in disease progression parameters. * Reduction in overall mortality.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection via investigator assessment, lab monitoring, and patient reporting.

Serious Adverse Events (SAEs): Standard 24-hour reporting timeline for unexpected or severe events. **Data Monitoring**

Committee (DMC): An unblinded independent review committee will assess safety data from the initial 60 participants (Cohort 1) before allowing enrollment into the broader Cohort 2.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy; Safety Set for all AE/SAE evaluations. **Sample Size Justification:** Target enrollment of up to 1,092 participants is designed to provide adequate statistical power ($\alpha = 0.05$, $Power = 80\%$) to detect meaningful differences in efficacy and safety between the two active doses of BMS-986278 and placebo. **Handling of Missing Data:** Standard handling methods for longitudinal pulmonary trial data (e.g., Mixed Model for Repeated Measures or Multiple Imputation for missing FVC data).

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring visits to ensure protocol adherence and data integrity. **Audit Trail:** Robust electronic Data Capture (eDC) mechanisms including eCRF locking and comprehensive data clarification processes.

11. Study Identifiers & Governance

Primary ID: IM027-1015 **Registry Number:** NCT06025578 **Sponsor/Lead Organization:** Bristol-Myers Squibb **Study Title:** ALOFT PPF Trial **Official Regulatory Title:** A Multicenter, Randomized, Double-blind, Placebo-controlled, Phase 3 Study to Evaluate the Efficacy, Safety, and Tolerability of BMS-986278 in Participants With Progressive Pulmonary Fibrosis

12. Clinical Framework (Progressive Pulmonary Fibrosis Specific)

Primary Condition: Progressive Pulmonary Fibrosis (PPF) **Diagnosis Threshold:** ILD with progressive features within 24 months + $\geq 10\%$ extent of fibrosis on HRCT **Medical Methodology:** Antifibrotic Treatment Pathway + Investigational LPA1 Antagonist **Optimization Target:** Slowing decline in lung function and delaying disease progression

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for Progressive ILD, allowing concomitant use of approved antifibrotics **Education Protocol (HCP):** Investigator training on protocol, spirometry standardization, and proper recording of HRCT scans **Education Protocol (Patient):** Protocol orientation, correct medication administration instructions (BID dosing), and reporting of adverse events **Quality Audit Mechanism:** Independent safety review committee (for Cohort 1) and centralized HRCT/Spirometry quality tracking

14. Observation & Follow-Up Schedule

Total Duration: Approximately 4 years (through late 2027) **Onsite Visit Cadence:** Regular intervals for safety labs and spirometry (e.g., Weeks 12, 24, 36, 52) **Remote Monitoring Frequency:** Between physical visits as dictated by standard PPF trial protocol **Checklist Review Interval:** Continuous throughout the study period

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				